Muhammad Aetzaz Ashraf

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Professional Summary

Software Engineer focused on AI/ML and NLP with hands-on experience building RAG, Agentic RAG systems, AI assistants, and intelligent automation workflows using Python, LangChain, LangGraph FastAPI, vector databases, and LLM integrations. Interested in developing practical AI applications for real-world use cases.

Technical Skills

Programming: Python, JavaScript, SQL

AI/ML: Machine Learning, NLP, RAG Pipelines

Frameworks & Libraries: FastAPI, LangChain, Scikit-learn, HuggingFace Transformers, Pandas, NumPy

LLMs & Vector DBs: Groq LLM, Qdrant, Embedding Models, BERT

Tools & Platforms: Docker, Git, GitHub, Postman, Kubernetes, GitHub Actions, MongoDb, SQL Server

Projects

Agentic Research Assistant with LangGraph RAG Pipeline 2026

- Built an Agentic RAG research assistant using LangGraph and LangChain with a multi-step workflow for retrieval, evaluation, and response generation.
- Designed a complete pipeline including document ingestion, text chunking, embedding generation, vector search, query routing, and LLM-based answer generation.
- Implemented intelligent routing and fallback workflows that evaluate retrieved context and trigger web search when relevant information is unavailable in the knowledge base.
- Integrated state management and modular AI nodes to support semantic search, query handling, and structured research responses.

AI-Powered RAG Document Assistant 2026

- Built a full-stack intelligent document assistant using Python (FastAPI) and LangChain that allows users to upload PDF, DOCX, CSV, and TXT files and receive AI-generated answers with citations.
- Implemented a complete Retrieval-Augmented Generation (RAG) pipeline with semantic search using Qdrant vector database and HuggingFace embeddings.
- Integrated Groq LLM with real-time streaming responses for fast and context-aware question answering.
- Designed scalable document ingestion, chunking, embedding generation, and retrieval workflows for efficient knowledge extraction.

Stock Sentiment Analysis using NLP & Machine Learning 2025

- Built an end-to-end NLP pipeline to analyze financial news headlines and predict stock market movement using TF-IDF, Logistic Regression, BERT embeddings, and XGBoost.
- Performed text preprocessing, feature engineering, sentiment analysis, and comparative model evaluation on real-world financial datasets.
- Implemented machine learning workflows for training, validation, and performance optimization of NLP models.
- Visualized insights and model performance metrics to compare traditional ML techniques with transformer-based embeddings.

Education

FAST-NUCES Islamabad, Pakistan

BS Software Engineering *June 2025*